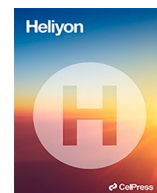




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Research article



Adsorptive removal of cadmium (II) from wastewater using activated carbon synthesized from stem of Khat (*Catha edulis*)

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ABSTRACT

Introduction: Cadmium is among the most hazardous heavy metals, posing the greatest risk to human beings and the environment. Adsorption with activated carbon prepared from agricultural waste is the most effective way to remove cadmium (II) from wastewater. In this study, activated carbon prepared from the stem of Khat (*Catha edulis*) plant was used for the removal of cadmium (II) from wastewater.

Objective: To evaluate the efficiency and mechanism of the removal of cadmium (II) from wastewater using Khat (*Catha edulis*) stem activated carbon at different operating parameters.

Method: Proximate and Fourier transform infrared analyses were conducted to characterise the prepared Khat stem-activated carbon. The effects of initial cadmium concentration, adsorbent dose, pH, contact time, and agitation speed on the cadmium removal efficiency of Khat stem-activated carbon were evaluated. Furthermore, isotherm and kinetic models of adsorption were used to evaluate the mechanism of cadmium removal.

Result: The proximate analysis indicated that the activated carbon derived from Khat stems possesses a bulk density of 0.58 g/cm³ and surface area of 615 m²/g. Additionally, the prepared Khat stem activated carbon has 4 %, 14 %, and 22 % of moisture content, ash content, and volatile matter, respectively. Furthermore, the proximate analysis indicated that the Khat stem activated carbon has a porosity of 55 %. The Fourier transformed infrared spectroscopy result indicated the presence of phenolic, alcoholic, and carboxylic acid functional groups on the surface of Khat stem activated carbon. The experimental data showed a better fit with the Langmuir isotherm model (R-squared and chi-square value of 0.9727 and 1.3936, respectively) and the pseudo-second-order kinetic model (R-squared and chi-square values of 0.9032 and 0.2179, respectively). The highest adsorption efficiency of cadmium (97 %) was attained at an adsorbent dose of 0.125 g, a contact

Abbreviations: KSAC, Khat stem activated carbon; Cd (II), cadmium (II) ion; FT-IR, Fourier transform infrared; PZC, point of zero charge; RPM, revolution per minute; AAS, Atomic Absorption Spectroscopy; ASTM, American Society for Testing Materials; Fig, Figure.

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time of 30 min, an initial cadmium concentration of 20 mg/L, a pH of 5, and an agitation speed of 100 rpm at room temperature (25 °C).

Conclusion: The results obtained in this research demonstrated that activated carbon from Khat stems can be employed as an economical, ecologically friendly, easily accessible and efficient activated carbon to remove Cd (II) from wastewater.

1. Introduction

The concentration of heavy metals in wastewater has been rising alongside industrial expansion and human activities. The prevalent heavy metals identified in industrial effluent comprise chromium, arsenic, lead, mercury, cadmium, copper, nickel, silver, and zinc [1,2]. Cadmium is a soft, silvery-white metal that mostly prefers a +2 oxidation state. It exists naturally in sedimentary rocks and the crust of the Earth at concentrations ranging from 0.1 to 15 mg/kg [3]. Along with lead, chromium, arsenic, and mercury cadmium is a hazardous metal that has no any role in human body [4]. Cadmium enters aquatic ecosystems via untreated industrial wastewater and is biomagnified further down the food chain [5]. The World Health Organization (WHO) considers cadmium as a major health risk [6]. According to Pure Earth's estimate, 5 million individuals worldwide are at risk of exposure to cadmium, and this exposure is expected to cause 250,000 years lost due to ill health, disability, or early death from disease [7]. Even in extremely small amounts, cadmium is harmful to humans. It accumulates in the kidneys, and it is a major cause of cancer and cardiovascular disease [6, 8]. It can also affect the neurological system, lungs, kidneys, liver, and reproductive organs in humans [5,9–11]. Aquatic species are also affected by cadmium. Due to its biological specificity, cadmium causes tissue variations and significant harm to the structure and function of tissues. It impairs the immune system, disrupts antioxidant defense mechanisms, and impairs reproductive functions [12]. Moreover, studies have revealed that in crop plants, cadmium toxicity inhibits the absorption and transport of water and nutrients. In addition, it impairs the physiology and structure of plants, causes oxidative damage, and interferes with metabolism [13,14]. About 25,000 tonnes of cadmium are discharged into the environment each year by natural processes. Around fifty percent of this amount is released into water bodies by the weathering of rocks, with the remaining amount being emitted into the atmosphere by volcanic eruptions and wildfires in forests. However, global annual cadmium production is approximately 14,000 tonnes. Canada is the leading producer, with other contributors including the Japan, Mexico, USA, Peru and USA [15]. A variety of anthropogenic sources, such as mining operations, industrial activities, and agricultural activities, contribute to cadmium emissions [4]. Nonetheless, a major source of cadmium emissions into the environment is the discharges of raw industrial effluent [16].

Wastewater discharged from different industries, including fertiliser and pesticide production, battery and plastics manufacturing, electroplating and metallurgical processes, mines, pigment and dye manufacturing, textile operations, and refining industries, is frequently contaminated with cadmium (II) [17–20]. In Ethiopia, the effluent from the garment and textile industries is the most significant contributor to cadmium pollution due to the use of cadmium for the production of color pigments in textile dyes [21]. Ethiopia currently has over 38 textile factories. Many of these factories do not have waste treatment facilities to properly treat cadmium (II) from wastewater [22]. They merely dispose of wastewater directly into the environment (mainly water bodies) without any treatment [23].

Previous studies conducted in Ethiopia have indicated that the concentration of cadmium in raw textile effluent reaches 0.18 mg/L [24] and that in raw paint effluent ranges from 1.58 to 69.22 mg/L [25]. These levels of cadmium are significantly greater than maximum discharge limits set by the World Health Organization and the USEPA (0.01 mg/L) [26]. Therefore, developing a realistic treatment option is essential for preventing the ecological and human health effects of cadmium pollution.

Previously, various methods, including ion exchange, membrane separation, and chemical precipitation, have been utilised all over the world to remove cadmium from industrial effluent. However, developing nations cannot afford these methods due to the generation of large amounts of sludge, the sensitive operating conditions, and high maintenance and operating costs [27]. Thus, research on low-cost agricultural waste by-products has become necessary due to the requirement for affordable and environmental friendly methods to remediate Cd (II) from cadmium contaminated effluent in recent years. As a result, adsorption using low-cost agricultural by-products is the most appropriate technique for low-income countries like Ethiopia [28]. The advantages of this approach include flexibility in design, high-quality treated effluent, low cost, high efficiency, low sludge production, adsorbent regeneration, and metal recovery [1].

Several studies have reported the removal of Cd (II) from industrial effluent utilising activated carbon prepared from different agricultural waste; these include pomegranate peel [29], corncob [30], bark of *Ficus racemosa* [31], rice husk [32], acacia senegal (L.) Willd pods [33], barley husks [34], wheat husks [35], coconut waste [36], and date pits [37]. In these studies, a maximum Cd (II) removal efficiency of 72 %–98.54 % was reported.

In the present investigation, activated carbon prepared from Khat (*Catha edulis*) stem was utilised to remove Cd (II) from wastewater. Khat is an evergreen dicotyledonous plant that exerts sedative effects on its leaves and is often used for chewing [38]. Ethiopia, Kenya, Somalia, Djibouti, Yemen, Eritrea, and Uganda are among the countries where it is frequently cultivated [39]. In Ethiopia, Khat stems are often discarded indiscriminately, particularly around Khat marketplaces, along roadsides, and near garbage dumpsters. This adds to the massive quantity of solid waste produced in Ethiopia's cities [38]. Furthermore, Khat stems have high cellulose, hemicellulose, and lignin contents [40]. These lignocellulose materials have a range of functional groups that aid in the complexation and adsorption of Cd (II) from the wastewater [41,42]. The Khat (*Catha edulis*) stem was selected for several reasons, including its low cost, easy accessibility, and environmental sustainability for making activated carbon.

The use of *Catha edulis* stem to prepare activated carbon for the adsorption of Cd (II) is innovative because there is insufficient information in the literature on its application for this purpose. Furthermore, to improve the removal efficiency of *Catha edulis* stem, the authors also made several adjustments to the activated carbon preparation process based on earlier research on other agro-waste-based activated carbon. To extent of the investigators' knowledge, no research has been done on the use of activated carbon made from the stems of Khat (*Catha edulis*) to remove cadmium (II) from wastewater. Therefore, in the present investigation, in more detail, the adsorption isotherms and kinetics, as well as the effects of variables such as contact time, initial concentration of cadmium (II), adsorbent dose, pH, and agitation speed, on the removal efficiency of Cd (II) were examined. Furthermore, the reusability of Khat stem activated carbon for cadmium (II) ion adsorption was studied by batch tests.

2. Materials and methods

2.1. Study design and period

An experimental study was carried out at Bahir Dar University, Debre Tabour University, and the University of Gondar from April 2022 to June 2022.

2.2. Chemicals and reagents

In this investigation, analytical-grade reagents and chemicals were used. The desired mass of $\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ was added to 1 L of distilled water to prepare a 1000 mg/L cadmium stock solution. The stock solution of Cd (II) was then diluted in distilled water to prepare working solutions of Cd (II) (20–100 mg/L). The pH of the working solution was adjusted using 0.1 M NaOH and 0.1 M HCl. Moreover, H_2SO_4 (98 %) and NaHCO_3 (2 %) were used during the preparation of KSAC.

2.3. Sample size determination and sampling procedure

Approximately 2 kg of Khat stem were collected in a pre-cleaned polyethylene bag from a Khat farm near Bahir Dar city using a simple random sampling method. The Khat stem were then taken to the Department of Chemistry Laboratory Facility at the University of Gondar for further experimental work. To indicate the effect of each predictor variable on the removal efficiency of Cd (II), a total of 30 synthetic samples were tested by selecting the values of the independent variables based on previous findings conducted using activated carbon made from other agricultural waste and by considering the characteristics of paint and textile factory wastewater in Ethiopia and the KSAC preparation conditions [29,43–47].

2.4. Adsorbent preparation

The KSAC preparation was conducted according to Ref. [48], with slight modification. Khat stem was collected from Khat farm, which is found around Bahir Dar city, using a pre-cleaned polyethylene bottle. The collected Khat stem underwent three rounds of washing in distilled water to remove any dirt or other contaminants. The stem was cut into pieces of 1–3 mm and dried in the oven at 105 °C for 24 h. The dried Khat stem was impregnated with 0.1 M H_2SO_4 (1:4 w/v) for 24 h. The acid-impregnated sample was neutralized by soaking in 2 % NaHCO_3 for 24 h, repeatedly rinsed with distilled water to remove any remaining acid residue, and filtered through Whatman filter paper 41. The resulting carbon was oven-dried at 105 °C for 24 h and activated in a furnace at 600 °C for 1 h. Finally, the KSAC was cooled to ambient temperature in a desiccator, sieved to the required particle size (250–500 µm), and preserved for further use.

2.5. Characterizations of Khat stem activated carbon

FT-IR-65 spectrometry was used to conduct an FT-IR analysis of the solid phase of the KSAC both before and after Cd (II) adsorption to determine the functional groups found on the surface of the KSAC. This analysis is based on changes in vibrational frequencies

Table 1
Standard methods used for the proximate analysis of Khat stem activated carbon.

Characteristics of KSAC	Standard methods used	Reference
Moisture content	ASTM D2867-99	[38]
Ash content	ASTM D2866-94	[49]
Volatile matter	ASTM D5832-95	[49]
Fixed carbon content	Previous study	[50]
Carbon yield	Previous study	[51]
pH	ASTM D3838-80 method	[52]
Point of zero charge	Solid addition method	[49]
Surface area	Sears method	[53]
Bulk density	ASTM D2854-96 method	[54]
Porosity	ASTM D2854-96 method	[51]

between 400 and 4000 cm^{-1} . Additionally, proximate analysis using ASTM (American Society for Testing and Materials) techniques was carried out for the prepared KSAC (Table 1). Finally, the Cd (II) removal efficiency of the KSAC was determined using (Eq. (1)).

$$\text{Removal efficiency (\%)} = \frac{C_0 - C_t}{C_0} \times 100 \quad 1$$

where C_0 (mg/L) is the initial cadmium (II) concentration and C_t (mg/L) is the cadmium (II) concentration at a time t .

2.6. Batch adsorption experiments

To evaluate the efficiency of KSAC in removing Cd (II) from aqueous solutions, batch adsorption studies were carried out using 100 mL conical flasks. Each flask was filled with 50 mL of various working solutions (20–100 mg/L). The experiments were carried out based on Adetokun et al. (2019) with a simple modification [33]. The Cd (II) stock solution was diluted with distilled water to obtain the desired concentrations of Cd (II) solutions. To adjust the pH of each working concentration of Cd (II), a 0.1 M solution of HCl or 0.1 M NaOH was used. Consequently, the desired mass of KSAC was also added, and the mixture was shaken at a specified agitation speed for a predetermined time at 25 °C. Finally, the mixtures were filtered using Whatman filter paper No. 41 and digested by adding 2 mL of HNO_3 until the volumes were reduced to 5 mL. The volume of each sample was then adjusted to 50 mL by adding distilled water, and the remaining concentration of Cd (II) in each sample mixture was measured by flame atomic absorption spectroscopy (AAS).

2.7. Effect of individual process parameters

The effect of individual parameters: pH [2,3,4,5, and 6], initial concentration of Cd (II) (20, 40, 60, 80, and 100 mg/L), adsorbent dose (0.025, 0.05, 0.1, 0.125, 0.15, and 0.175 g), contact time (5, 10, 15, 20, 30, 40, 50, and 60 min), and agitation speed (50, 100, 150, 200, and 250 rpm) on Cd (II) removal efficiency from wastewater were evaluated at room temperature (25 °C) in a 50 mL Cd (II) solution. While evaluating the effect of each parameter, the other 4 parameters were set to a constant value, and the constant values were generated from previous studies conducted on agro-waste-based activated carbon, which were pH: 5, initial concentration of Cd (II): 20 mg/L, adsorbent dose: 0.125 g, contact time: 30 min, and agitation speed of 100 rpm.

3. Adsorption mechanism study

3.1. Adsorption isotherm

The Freundlich and Langmuir isotherm models (Eqs. (2) and (4)) were assessed to determine the mechanisms of interaction between Cd (II) and the KSAC. Due to its empirical nature and because it takes into account the existence of a structure with multiple layers, the Freundlich isotherm offers the most comprehensive model of adsorption on heterogeneous surfaces. In contrast the Langmuir model, anticipates the creation of an adsorbed solute monolayer with no interactions between the adsorbed Cd (II) ions and expects that one Cd (II) ion will adsorb on each binding site on the homogeneous surface of the adsorbent [55]. The Langmuir isotherm model in its non-linear form is expressed by Eq. (2) [56].

$$q_e = \frac{q_{\text{max}} \cdot b \cdot C_e}{1 + b \cdot C_e} \quad 2$$

where C_e (mg/L) represents the concentration of Cd (II) at equilibrium, q_e (mg/g) denotes the amount of Cd (II) adsorbed at equilibrium, q_{max} (mg/g) is the KSAC saturation capacity, and b (L/mg) is the adsorption constant. The Cd (II) uptake at equilibrium for isotherm determination (q_e) is described using Eq. (3) [56].

$$q_e = \frac{(C_0 - C_e) \cdot V}{m} \quad 3$$

where q_e (mg/g) represents the amount of Cd (II) adsorbed, C_0 (mg/L) denotes the initial Cd (II) concentration in the solution, C_e (mg/L) refers to the remaining Cd (II) concentration at equilibrium, V (L) is the solution volume in the batch, and m (g) is the mass of KSAC used.

The non-linear Freundlich isotherm model is described by Eq. (4) [56].

$$q_e = K_f \cdot C_e^{1/n} \quad 4$$

where the adsorption intensity and capacity are denoted by the Freundlich constants n and K_f , respectively.

3.2. Adsorption kinetics study

The pseudo-first-order and pseudo-second-order kinetic models were assessed to evaluate the adsorption rate of Cd (II) by KSAC. The pseudo-first-order kinetics model in its non-linear form is evaluated using Eq. (5) [56].

$$qt = qe(1 - e^{-k_1 t}) \quad 5$$

where q_e (mg/g) is the quantity of Cd (II) adsorbed per gram of the KSAC at equilibrium, q_t (mg/g) is the quantity of Cd (II) adsorbed per gram of the KSAC at time t , and $K_1(\text{min}^{-1})$ refers to the pseudo-first-order kinetic rate constant.

The pseudo-second-order kinetic model in its non-linear form is evaluated using Eq. (6) [56].

$$qt = \frac{k_2 * qeq^2 * t}{1 + k_2 * qeq * t} \quad 6$$

where q_t (calculated using Eq. (7)) and q_{eq} represent the quantity of Cd (II) adsorbed at time t and equilibrium, respectively, and k_2 is the rate constant of the pseudo-second-order kinetic.

$$qt = \frac{(C_0 - C_t) * V}{m} \quad 7$$

where C_t (mg/L) is the remaining amount of Cd (II) at a time t .

3.3. Data quality control

Proper handling and preservation of equipment were conducted to prevent cross-contamination. The analysis of each sample was done in triplicate. In addition, analytical-grade chemicals were used. Moreover, instrumental calibration was conducted using standard concentrations of Cd (II) (0.05, 0.075, 1, 2.5, 5, 7.5, and 10 mg/L). Sample analysis was carried out when the R-squared value exceeded 0.99, based on the calibration curve.

4. Result and discussion

4.1. Preparation of the adsorbent

The KSAC was prepared based on previous studies on other agricultural waste-activated carbon with little modification (Fig. 1). [48].

4.2. Proximate analysis of the KSAC

According to its physicochemical characteristics, the generated KSAC is of sufficiently high grade for cadmium (II) removal (Table 2).

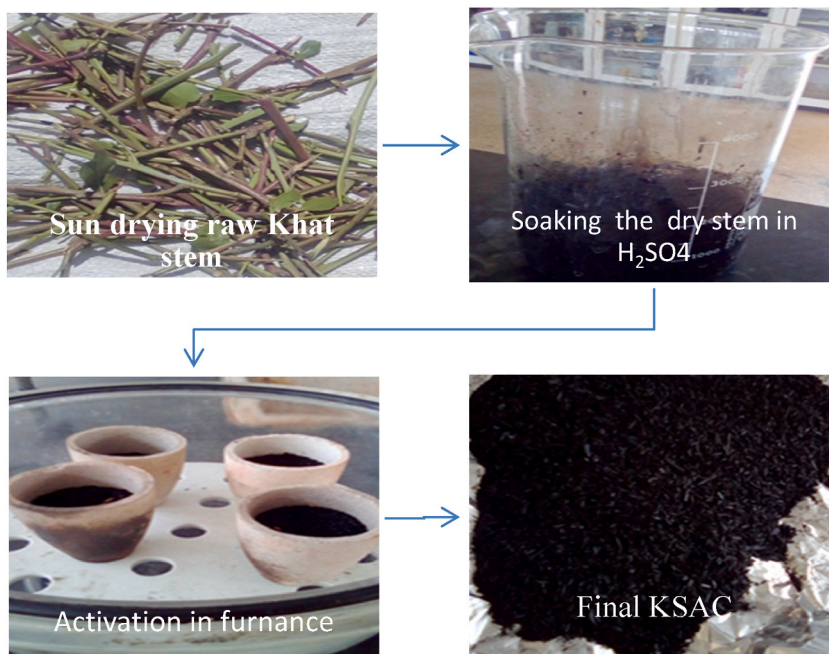


Fig. 1. Different stages of Khat stem activated carbon production.

As clearly indicated in Table 2, the moisture and ash contents of the KSAC were 4 % and 14 %, respectively, indicating that the KSAC has excellent quality. The low moisture and ash content could be due to the release of water and mineral content during the carbonisation and activation processes of the KSAC preparation [57]. As shown in Table 3, the KSAC prepared in this study contains a similar moisture content to that of sawdust activated carbon [58]. In addition, the KSAC contains lower moisture and ash content than activated carbon prepared from Parthenium hysterophorus weed and rice husk [59,60]; however, the KSAC contains higher moisture and ash content than activated carbon prepared from coconut shells [61]. The amount of volatile matter and fixed carbon in activated carbon serve as markers for the development of porosity and degree of carbonisation [62]. In the present study, the produced KSAC contains relatively low volatile matter and high fixed carbon content than activated carbon prepared from rice husk and wheat straw [63]. However, the KSAC contains lower fixed carbon and higher volatile matter content than Prosopis juliflora activated carbon [64].

A high surface area of an adsorbent is preferred for wastewater treatment applications because it increases the amount of adsorbate adsorbed on its surface [68,69]. The surface area of activated carbon typically used for commercial applications generally falls between 500 and 1500 m²/g, and can occasionally reach up to 3000 m²/g [70]. The current investigation is consistent with the aforementioned values, having a surface area of 615 m²/g. This could be due to carbonisation and activation processes, in which volatile compounds are released as gases and make organic substances unstable due to heat, breaking down molecules' bonds and links [71]. The surface area of the KSAC prepared in this study is comparable to that of rice-husk activated carbon [60]. However, it is higher than sugarcane bagasses activated carbon [72] and Noug stalk activated carbon [69].

The high carbon yield of the KSAC, as reported in Table 2, indicates the feasibility of the Khat stem as a raw material for the preparation of activated carbon [66]. The KSAC in this study contains a higher carbon yield than other agro-wastes such as rice husk and coffee husk [73,65]. The bulk density of activated carbon must be greater than 0.25 g/cm³ to be used effectively [74]; in the current investigation, the bulk density of the KSAC was reported to be 0.58 g/cm³, which is higher than the aforementioned requirements. However, the bulk density in this study is lower than that of rice husk activated carbon [60].

High porosity of activated carbon is preferred in the treatment of wastewater because its adsorptive characteristics derive from its extensive internal pore structures [75]. The porosity of the KSAC in this study was found to be 55 % as shown in Table 3, which is greater than that of Teff husk activated carbon [66], but it is lower than that of Noug stalk activated carbon [67]. The pH^{pzc} value of the KSAC was 3.7. Thus, pH < pH^{pzc} suggests that the KSAC surface carries a net positive charge, while pH > pH^{pzc} suggests that the KSAC surface carries a net negative charge. The lower pH^{pzc} of this study could be due to the presence of acid functional groups on the KSAC surface, which might be attributed to H₂SO₄ activation [76].

4.3. Fourier transform infrared spectrophotometer (FTIR) test

Using the Fourier Transform Infrared Spectrometer 65, FT-IR analysis in solid phase was carried out on the KSAC both before and after Cd (II) removal to assess the KSAC's efficiency for Cd (II) removal from wastewater. The functional groups associated with each peak and stretches are displayed in Figs. 2 and 3. Peaks shifted from 3423 cm⁻¹ to 3416 cm⁻¹, 2918 cm⁻¹ to 2913 cm⁻¹, 1627 cm⁻¹ to 1620 cm⁻¹ and 1109 cm⁻¹ to 1102 cm⁻¹ are clearly shown before and after adsorption of cadmium (II), respectively. The interaction between cadmium (II) and -OH groups of the KSAC causes the distinctive adsorption peak of the O-H stretching vibration to shift from 3423 cm⁻¹ to 3416 cm⁻¹, indicating a reduction in the free hydroxyl group concentration. Furthermore, Figs. 2 and 3 indicates that the C-H, C=C and -C-O- functional groups are also present on the KSAC surface. This implies that the phenolic, alcoholic, or carboxylic acid groups of the KSAC might significantly influence the adsorption of Cd (II). The findings of this study are supported by previous studies conducted on activated carbons derived from agricultural waste [64,67].

5. Batch adsorption experiment

5.1. The effect of initial Cd (II) concentration on cadmium (II) removal efficiency

As clearly indicated in Fig. 4, the KSAC efficiency for Cd (II) removal was sensitive to changes in cadmium (II) concentration. As the initial concentration of Cd (II) increased, its removal efficiency declined. At a cadmium (II) concentration of 20 mg/L, the highest Cd (II) removal efficiency of 97 % was obtained. This might be because of the presence of many active sites at low initial Cd (II)

Table 2
Proximate analysis of Khat stem activated carbon.

Characteristics of KSAC	Units	Result
Moisture content	%	4
Ash content	%	14
Volatile matter	%	22
Fixed carbon content	%	60
Carbon yield	%	54
pH	–	5
Point of zero charge	–	3.7
Surface area	m ² /g	615
Bulk density	g/cm ³	0.58
Porosity	%	55

Table 3

Comparison of the characteristics of Khat stem activated carbon with activated carbon prepared from different agro-wastes.

Activated carbon	Characteristics of the activated carbon									Reference
	Moisture Content (%)	Ash Content (%)	Volatile Matter (%)	Fixed Carbon (%)	Carbon yield (%)	Point of zero Charge	Surface area (m ² /g)	Bulk Density (g/cm ³)	Porosity (%)	
Khat stem	4	14	22	60	54	2	615	0.58	0.55	This study [59].
Parthenium hysterophorus weed	5	19	9.5	66	–	–	256.5	–	–	
Coconut shell	2	10.7	–	–	–	–	936	0.5	–	[61].
Rice husk	3.8	2.3	53	41	34.1	–	281.7	–	–	[63].
Wheat straw	5.7	1	38	55.4	31.2	–	282.1	–	–	[63].
Coffee husk	6.05	2.48	28.46	–	43.3	–	–	–	–	[65].
Saw dust	4	2.5	30.2	–	–	–	–	–	–	[58].
Rice husk	17.72	18.45	58.007	–	–	5.1	615.2	0.88	–	[60].
Prosopis juliflora	2.98	0.92	17.54	78.56	–	–	600.4	–	–	[64].
Teff husk	10.1	17.5	21.2	51.2	58.42	5.1	327	0.549	42.5	[66].
Noug stalk	6	8	31	55	53.78	5.43	473.45	0.37	67.5	[67].

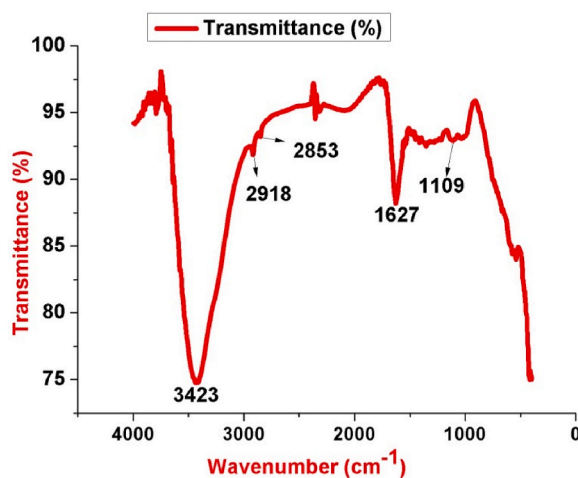


Fig. 2. Fourier transformed infrared analysis of Khat stem activated carbon before adsorption of cadmium (II).

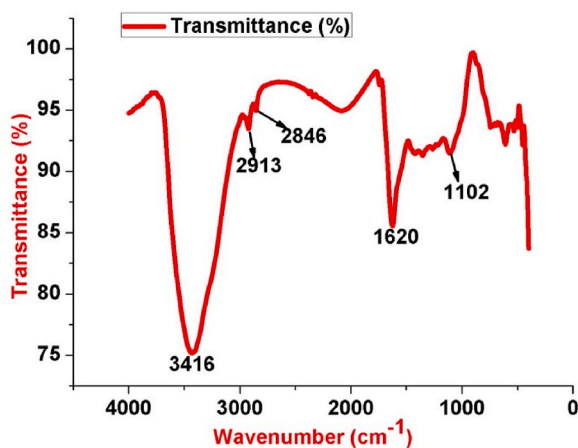


Fig. 3. Fourier Transform Infrared analysis of Khat stem activated carbon after adsorption of cadmium (II).

concentrations [77]. The findings of this study are supported by other studies [76,78].

5.2. The effect of bio sorbent dose on cadmium (II) removal efficiency

The results displayed in Fig. 5 indicate that when the dose of the KSAC was increased from 0.025 g to 0.150 g, the removal efficiency of Cd (II) also increased. However, the Cd (II) removal efficiency was slightly reduced when the KSAC dose was increased further, from 0.150 g to 0.175 g. This could be because a high dose of the KSAC up to the optimal value increases the KSAC's surface area, which increases the total number of active sites that are available for the adsorption of Cd (II) [79,80]. However, further addition of the KSAC beyond the optimum dose would cause particle interaction, such as aggregation, which reduces the KSAC's overall surface area and lowers its cadmium (II) adsorption efficiency [79]. The results of this study are supported by similar studies on the use of activated carbon derived from agricultural waste to remove Cd (II) from wastewater [79,81].

5.3. The effect of pH on Cd (II) removal efficiency

The KSAC's point of zero charge (pH^{PZC}) in this study was 3.7. Therefore, the surface of the KSAC is anticipated to be negative at $\text{pH} > \text{pH}^{\text{PZC}}$ and positive at $\text{pH} < \text{pH}^{\text{PZC}}$ [82]. The results of the batch adsorption experiments (Fig. 6) indicate that the adsorption efficiency of KSAC significantly enhanced as the pH of the solution increased from 4 to 5. The low removal efficiency of Cd (II) at lower pH is attributed to the positive charges on the KSAC surface when $\text{pH} < \text{pH}^{\text{PZC}}$. Conversely, when $\text{pH} > \text{pH}^{\text{PZC}}$, the cadmium removal efficiency of KSAC improved due to the negatively charged surface, which facilitated cadmium adsorption until the KSAC became saturated [83]. However, further increasing the pH above its optimum value decreases the efficiency of Cd (II) removal. This might be due to supersaturation of the hydroxyl ions (OH^-) in the solution [84]. At pH 5, the highest cadmium (II) removal efficiency was obtained. The results of this study are also consistent with those of previous studies [31,85].

5.4. The effect of contact time on cadmium removal efficiency

As clearly indicated in Fig. 7, the Cd (II) removal efficiency increased with increasing contact time. This could be because of the presence of several active binding sites early in the process of adsorption; however, as time passes and the active binding sites become filled with cadmium (II) ions, the process becomes slow [80]. A 60-min contact time resulted in the highest cadmium adsorption efficiency of 96.5 %. The findings of the present investigation are in line with those of previous investigation done on activated carbon produced from agricultural waste for Cd (II) removal [30,86].

5.5. The effect of agitation speed

The findings of this study, as indicated in Fig. 8, indicate that while the agitation speed increased to its optimum value, the removal efficiency of cadmium (II) increases; however, further increments of the agitation speed from its optimum value decreased the cadmium (II) removal efficiency. This might be due to the increased turbulence in the liquid phase, which increased the external mass transfer coefficient and reduced the boundary layer effect. This led to faster adsorption as the external resistance decreased [87]. However, further increments of the agitation speed from the optimal value lead to higher kinetic energy for cadmium (II) and adsorbent particles, resulting in the detachment of loosely bound cadmium (II) ions [87]. The highest Cd (II) removal efficiency was obtained at an agitation speed of 100 rpm. The findings of the present study are in line with those of previous studies on activated

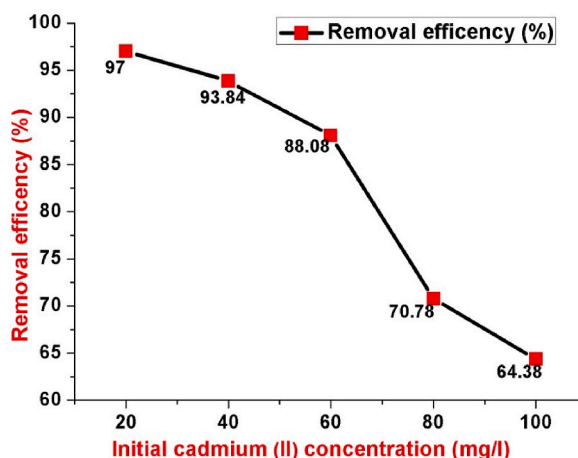


Fig. 4. Effect of the initial concentration of Cd (II) on the removal efficiency of Cd (II) under optimum adsorbent dose of 0.125 g, pH 5, contact time of 30 min, and agitation speed of 100 rpm at room temperature in a 50 mL Cd (II) solution.

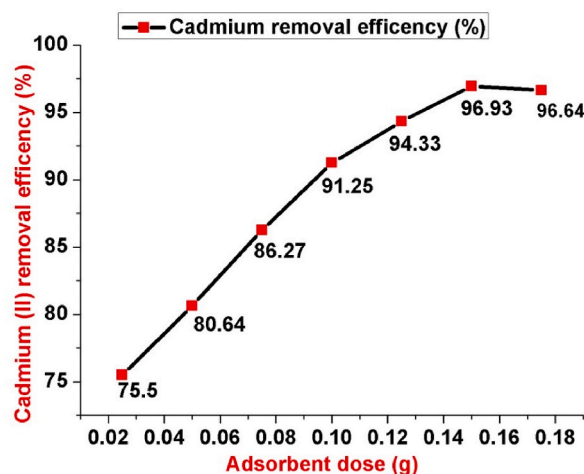


Fig. 5. Effect of adsorbent dose on the removal efficiency of Cd (II) under optimum initial concentration of cadmium (II) of 20 mg/L, pH 5, contact time of 30 min, and agitation speed of 100 rpm at room temperature in a 50 mL Cd (II) solution.

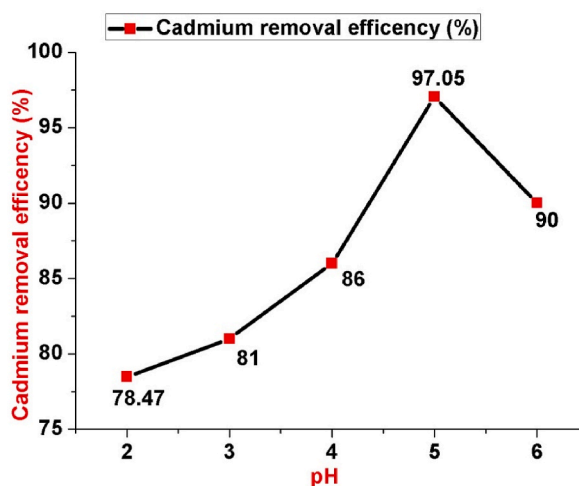


Fig. 6. Effect of pH on removal efficiency of Cd (II) under optimum adsorbent dose of 0.125 g, contact time of 30 min, initial concentration of Cd (II) of 20 mg/L, and agitation speed of 100 rpm at room temperature in a 50 mL Cd (II) solution.

carbon produced from agricultural waste for the removal of Cd (II) [88–90].

5.6. Comparing the cadmium (II) removal efficiency of KSAC with other activated carbons

The results in Table 4 compare KSAC's Cd (II) removal efficiency with those of other studies. In this study, a highest cadmium removal efficiency of 97 % was achieved at an adsorbent dose of 0.125 g, a contact time of 30 min, an initial cadmium concentration of 20 mg/L, a pH of 5, and an agitation speed of 100 rpm at 25 °C. The cadmium removal efficiency in this study is in line with previous studies [64,91]. However, the cadmium (II) removal efficiency of KSAC was higher than that of other ACs such as rice husk [32], pomegranate husk [92], and walnut shell-rice husk [93]. Moreover, the removal efficiency was lower than that reported in similar studies on *Acacia senegal* (L.) willd pods [33] and sunflower stems [94].

6. Adsorption mechanism study

6.1. Adsorption isotherm analysis

The isotherm curves provide insights into the maximum adsorption capacity of KSAC and the mechanisms of interaction between Cd (II) and KSAC [56]. In the present investigation, the adsorption isotherm was examined at an initial Cd (II) concentration of 20, 40,

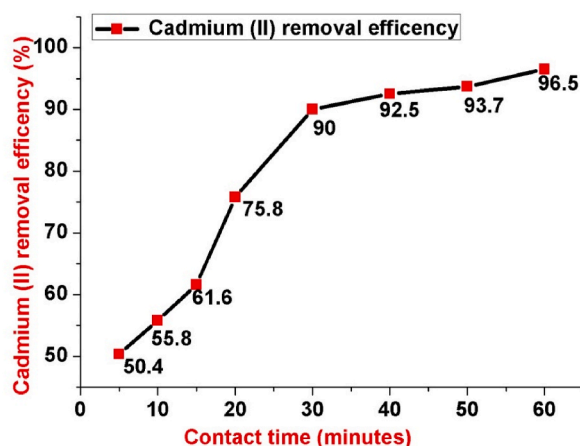


Fig. 7. The effect of contact time on the removal efficiency of Cd (II) under an optimum adsorbent dose of 0.125 g, a pH of 5, an initial cadmium concentration of 20 mg/L, and an agitation speed of 100 rpm at room temperature in a 50 mL Cd (II) solution.

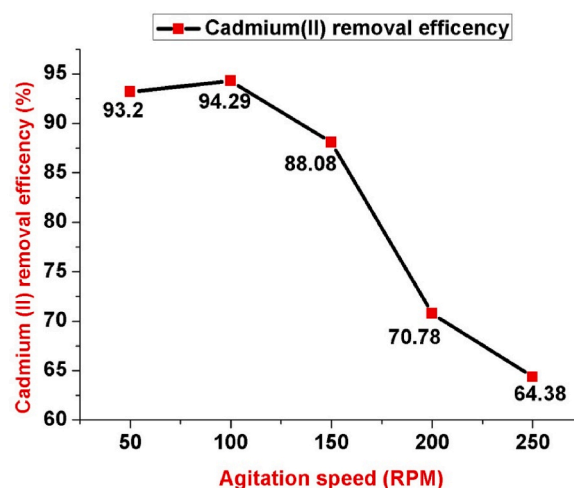


Fig. 8. The effect of agitation speed on the removal efficiency of Cd (II) under an optimum adsorbent dose of 0.125 g, contact time of 30 min, initial concentration of Cd (II) 20 mg/L, and pH of 5 at room temperature in a 50 mL Cd (II) solution.

Table 4

Comparison of cadmium (II) removal efficiency of Khat stem activated carbon with other adsorbents.

Activated carbon	Maximum Cadmium (II) removal efficiency (%)	Reference
KSAC	97	This study
Pomegranate husk	92.85	[92]
Walnut shell-rice husk	78.58	[93]
Padina gymnospora	96.2	[91]
Prosopis juliflora:	96.7	[64]
Rice husk	72	[32]
Acacia senegal (L.) Willd pods	98.54	[33]
Sunflower stem	100	[94]

60, 80, and 100 mg/L, with an adsorbent dose of 125 mg, pH of 5, contact time of 30 min, and agitation speed of 100 rpm in a 50 mL Cd (II) solution at room temperature (25 °C). According to the values of the correlation coefficients ($R^2 = 0.9727$) and chi-square (1.3936) obtained, the Langmuir model (Fig. 9) was better fitted to the Cd (II) removal experimental isotherm than the Freundlich isotherm (Fig. 10) with an R^2 and chi-square values of 0.8856 and 5.838, respectively. Furthermore, the maximum adsorption capacity (q_{max}) of KSAC was found to be 25.56 mg/g (Table 5). This would indicate that the process is carried out in a monolayer; for every site, the adsorption energy is the same. Adsorbate does not transmute in the plane of the surface. Once a pollutant occupies a site, more

adsorption cannot occur there, and molecules adsorbed on nearby sites do not interact with one another [56]. The results of this study are consistent with those of other studies that investigated the Cd (II) adsorption isotherm on activated carbon derived from agricultural waste [56,95].

6.2. Adsorption kinetics analysis

The kinetic models of adsorption provide an understanding of the reaction pathways and mechanisms [77]. In this study, the kinetic experiments were conducted at constant metal ion concentration (20 mg/L), adsorbent dose (3 g/L), pH [5], and agitation speed (100 rpm) over various contact times (5, 10, 15, 20, 30, 40, 50, and 60 min), and at room temperature (25 °C). To evaluate the adsorption mechanism, the pseudo-first-order kinetic model (Eq. (5)) and the pseudo-second-order kinetic model (Eq. (6)) were applied. Figs. 11 and 12 show the experimental kinetic curves for the removal of Cd (II) on the KSAC and the corresponding adjustments to the pseudo-first-order and pseudo-second-order kinetic models, respectively. Based on the values of the correlation coefficients ($R^2 = 0.9032$) and chi-square (0.2179), the pseudo-second-order model was better fitted to the Cd (II) removal experimental kinetics than the pseudo-first-order kinetics with correlation coefficients ($R^2 = 0.8355$) and chi-square (0.3704) (Table 6). This indicates that the adsorption might be chemical due to the covalent bonds between cadmium (II) and KSAC [96]. The findings of the present study are consistent with those of previous studies [92,97].

6.3. Intra-particle diffusion study

We have used the intra-particle diffusion model to forecast the rate-limiting steps as well as to determine the reaction routes and mechanisms for the adsorption of Cd (II) from wastewater using the KSAC [98]. The model was determined using Weber and Morris equation (Eq. (8)) [99].

$$qt = kd^*t^{0.5} + c \quad 8$$

where qt (mg/g) is the quantity of Cd (II) adsorbed at time t , kd (mg/g min) is the rate constant of the intra-particle diffusion model, which can be determined from the slope of qt versus $t^{0.5}$ plot. Additionally, the constant c (mg/g), which is proportional to the boundary layer's thickness, can be derived from the intercept of the same plot.

As shown in Fig. 13, the linear plot of qt versus $t^{0.5}$ did not intersect the origin, and the points were dispersed across the graph because of variations in the mass transfer rates during the first and last stages of adsorption. This implies that intra-particle diffusion is not the sole rate-limiting step in the adsorption. The observed deviation indicates multilinearity, meaning that the movement of the Cd (II) from the solution to the surface of the KSAC is governed by the simultaneous occurrence of multiple steps, such as film/external diffusion, surface diffusion, and adsorption onto the surface of pores [100].

6.4. Thermodynamic studies

Thermodynamic parameters provide valuable insights into the explanation of adsorption behaviour. A thermodynamic analysis was conducted to assess the spontaneity of the study at 25 °C (298 K). The Gibbs energy change (ΔG°) was determined as follows using Eq. (9) [55].

$$\Delta G^\circ = -RT \ln K \quad 9$$

where T represents the temperature in Kelvin, R stands for the gas constant ($8.314 \text{ J mol}^{-1}\text{K}^{-1}$), and K refers to the thermodynamic equilibrium constant, which was calculated from the Langmuir isotherm constant by applying Eq. (10) [101].

$$K = \frac{(1000 \cdot b \cdot mw) \cdot A}{\gamma} \quad 10$$

where b represents the Langmuir isotherm constant (L/mg), mw (mg/mol) is the molar mass of cadmium, A is the standard concentration of the adsorbate (1 mol L^{-1}), and γ is the activity coefficient (dimensionless), which approaches unitary value for the dilute concentration of the adsorbate.

The calculated value of ΔG° was -27.699 kJ/mol at 298 K, indicating that the adsorption of cadmium (II) by the KSAC was feasible and spontaneous.

6.5. Desorption studies

Regeneration of the KSAC was performed by diluting the dissolved KSAC in 1 M HNO_3 for 1 h. After that, the KSAC was filtered, rinsed with distilled water, and oven-dried at 105 °C for 24 h. Lastly, the regenerated KSAC was reapplied to the new cadmium ion solution at an adsorbent dose of 0.125 g, a contact time of 30 min, an initial cadmium concentration of 20 mg/L, an agitation speed of 100 rpm, and a temperature of 25 °C. The reusability of the KSAC is summarized in Table 7. The percentage of cadmium ion removal decreased between the KSAC's first, second, third, and fourth reuse cycles. This might be because of decreased site activity and the blockage of KSAC pores [55]. From these results, we concluded that protons displace large amounts of adsorbed cadmium in acidic

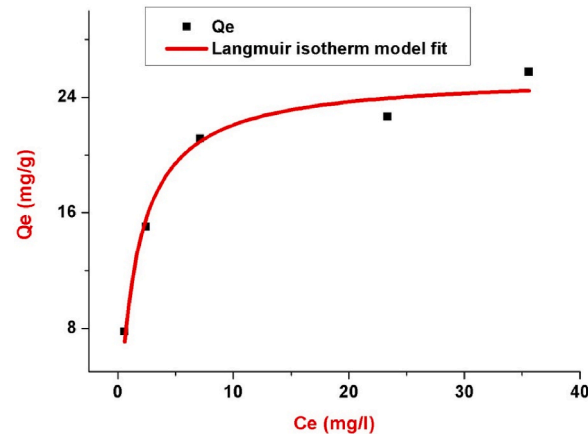


Fig. 9. Plot of Langmuir isotherm at initial Cd (II) concentrations of 20 mg/L, 40 mg/L, 60 mg/L, 80 mg/L, and 100 mg/L, adsorbent dose of 125 mg, pH of 5, contact time of 30 min, and agitation speed of 100 rpm at room temperature in a 50 mL Cd (II) solution.

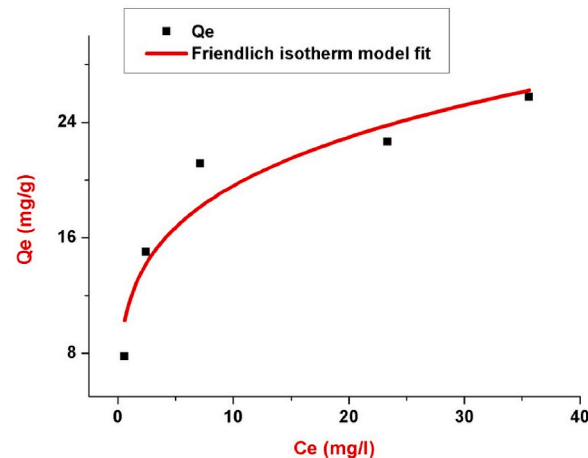


Fig. 10. Plot of Freundlich isotherm at initial Cd (II) concentrations of 20 mg/L, 40 mg/L, 60 mg/L, 80 mg/L, and 100 mg/L, adsorbent dose of 125 mg, pH of 5, contact time of 30 min, and agitation speed of 100 rpm at room temperature in a 50 mL Cd (II) solution.

Table 5
Parameters for nonlinear Langmuir and Freundlich isotherm model.

Model	Parameter	Value
Non-linear Langmuir isotherm	qmax	25.54598
	b	0.63766
	R ²	0.97271
	X ² (chi-square)	1.39368
Non -linear Freundlich isotherm	Kf	11.56056
	1/n	0.2297
	R ²	0.88566
	X ² (chi-square)	5.83879

media by competing with Cd (II) ions for adsorption sites [46]. The results of this study are in line with those of other previous studies [33,102].

6.6. Final fate of cadmium (II) adsorbed KSAC

Because cadmium is a non-renewable resource, it can be recycled for industrial purposes. Therefore, cadmium will be extracted and reintroduced into the industrial sector.

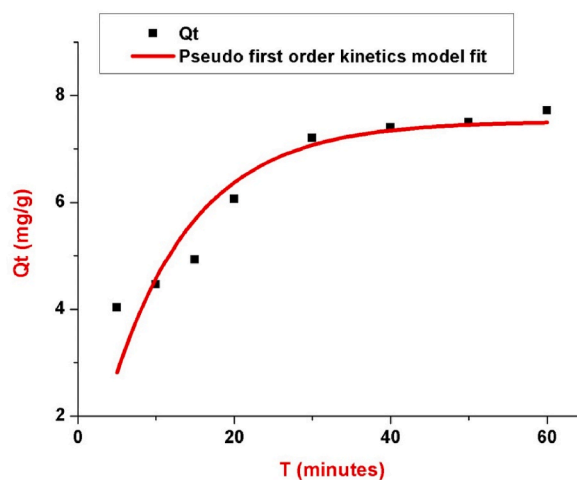


Fig. 11. Plot of Pseudo first order kinetics model at contact time of (5, 10, 15, 20, 30, 40, 50, 60 min), initial Cd (II) concentration (20 mg/L), adsorbent dose (125 mg), pH [5], agitation speed (100 rpm) at room temperature in 50 mL Cd (II) solution.

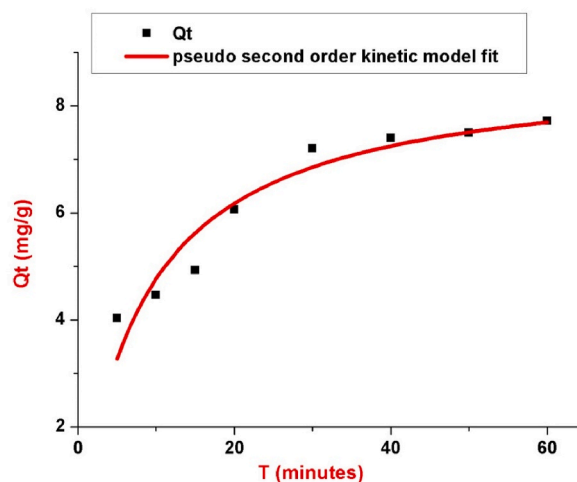


Fig. 12. Plot of pseudo-second-order kinetic model at contact time (5, 10, 15, 20, 30, 40, 50, 60 min), initial Cd (II) concentration (20 mg/L), adsorbent dose (125 mg), pH [5], agitation speed (100 rpm), at room temperature in 50 mL of Cd(II) solution.

Table 6

Parameters for nonlinear pseudo-first order and pseudo-second order kinetics model.

Model	Parameter	Value
Nonlinear pseudo first order	qe	7.52118
	K1	0.09386
	R2	0.83556
	X ² (chi square)	0.37044
Nonlinear pseudo second order	K2	0.0136
	qeq	8.76765
	R ²	0.90325
	X ² (chi-square)	0.21795

6.7. Health implication

Cadmium enters aquatic ecosystems via the discharge of untreated industrial effluents, is biomagnified farther down the food chain, and poses a public health concern [5]. Various investigations have assessed the risks to the general public's health associated with exposure to cadmium by multiple routes, such as drinking water, inhaling dust, skin contact with soil and water, and eating food

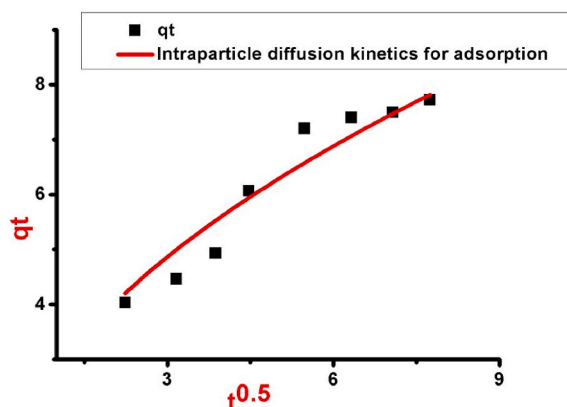


Fig. 13. The intraparticle diffusion kinetics for the removal of Cd (II) at an initial Cd (II) concentration (20 mg/L), adsorbent dose (125 mg), pH [5], and agitation speed (100 rpm), at room temperature.

Table 7

Reuse assessment of Khat stem activated carbon at an adsorbent dose of 0.125 g, a contact time of 30 min, an initial cadmium concentration of 20 mg/L, and an agitation speed of 100 rpm at room temperature (25 °C) in a 50 mL Cd (II) solution.

Cycle of reuse	Cadmium removal efficiency (%)
1st	97
2nd	74.50
3rd	63.7
4th	45

grown in polluted fields [103]. When crops such as rice, wheat, potatoes, and other vegetables are grown in cadmium-polluted water due to industrial discharge, high concentrations of cadmium accumulate in them, resulting in various health risks [19]. For example, a study conducted in Pakistan indicated potential health risks from the accumulation of cadmium in food crops grown in cadmium-polluted water [104]. Another study conducted in Bangladesh indicated the presence of significant health risks from drinking water contaminated with cadmium from textile industries [20]. Similarly, a study conducted in Ethiopia indicated that elevated level of cadmium in labeobarbus intermedius fish from cadmium-contaminated rivers could pose a risk of developing cancer in the future [105]. In this study, the prepared Khat stem activated carbon removed 97 % of the cadmium (II) from wastewater, implying that the WHO permissible limits for the discharge of industrial wastewater for cadmium [26] can be achieved for some types of industrial wastewater, and as a result, the health risks from industrial cadmium (II) pollution can be minimised.

6.8. Limitation of the study

A limitation of the present study is that the impacts of temperature, particle size, and other competing cations on the Cd (II) removal efficiency of KSAC were not investigated due to limited resources. Furthermore, a detailed morphological analysis of the KSAC was not conducted.

7. Conclusion and recommendation

This work developed activated carbon from the stem of Khat plant to remove cadmium (II) from wastewater. The synthesized KSAC was characterised using FTIR and proximate analysis. The proximate analysis indicated that the KSAC has a surface area of 615 m²/g and a porosity of 55 %, while the FTIR analysis indicated the presence of phenolic, alcoholic, and carboxylic acid functional groups on the KSAC's surface. The batch adsorption experiments showed that the highest percent removal of cadmium (97 %) was achieved at an adsorbent dose of 0.125 g, a contact time of 30 min, an initial cadmium concentration of 20 mg/L, a pH of 5, and an agitation speed of 100 rpm. The experimental data suited the Langmuir isotherm better, indicating that the adsorption mechanism was a monolayer process. Furthermore, the pseudo-second-order kinetic model was found to be the better fitting adsorption kinetic model, confirming the role of chemisorption in the adsorption process. Although the reuse assessment indicates a decrease in the efficiency of the KSAC for the adsorption of Cd (II) in each consecutive cycle, the regenerated KSAC can still be reused up to the 3rd cycle. Therefore, KSAC can be employed as an economical, ecologically friendly, easily accessible, and efficient activated carbon to remove Cd (II) from wastewater.

CRediT authorship contribution statement

Yenewa Bewket Gelaw: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Henok Dagne:** Supervision. **Balew Adane:** Writing – review & editing. **Getasew Yirdaw:** Writing – review & editing. **Mekonnen Moges:** Writing – review & editing. **Zelalem Aneley:** Writing – review & editing. **Lake Kumlachew:** Writing – review & editing. **Abebaw Aschale:** Writing – review & editing. **Yikeber Argachew Deml:** Writing – review & editing. **Eniyew Tegegne:** Writing – review & editing. **Tsegaye Adane Birhan:** Supervision, Methodology, Formal analysis.

Consent to publish

Not applicable.

Ethical approval

Ethics approval was received from the ethics review committee of the Institute of Public Health at the University of Gondar. Verbal consent was also received from the Amhara Regional State Environment, Forestry, and Wildlife Protection Bureau for collecting Khat stem. Furthermore, verbal consent was obtained from the owner of the Khat farm for collecting the Khat stem samples.

Data availability

All data used in this study is available from the corresponding author upon reasonable request.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Corrigendum

Corrigendum to “Adsorptive removal of cadmium (II) from wastewater using activated carbon synthesized from stem of Khat (*Catha edulis*)” [Volume 10, Issue 22, 30 November 2024, e40389]

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In this article, the authors have included reference [54] and [71], which requires correction due to contextual inappropriateness. The authors confirm that these references were inadvertently introduced in the article, and the updated references reflect the accuracy of the cited sources.

The original reference and its associated citation in the sections "2.5. Characterizations of Khat stem activated carbon (Table 1)" and "4.2. Proximate analysis of the KSAC" can be found below:

[54] A. Dada, A. Olalekan, A. Olatunya, O. Dada, Freundlich Langmuir. Temkin and Dubinin–Radushkevich isotherms studies of equilibrium sorption of Zn²⁺ onto phosphoric acid modified rice husk. IOSR J. Appl. Chem., 3 (1) (2012), pp. 38-45

2.5. Characterizations of Khat stem activated carbon

Table 1. Standard methods used for the proximate analysis of Khat stem activated carbon.

Bulk density ASTM D2854-96 method [54]

[71] J. Speight. Production of fuels from nonfossil fuel feedstocks. The Refinery of the Future (2020), pp. 391-426

4.2. Proximate analysis of the KSAC

The current investigation is consistent with the aforementioned values, having a surface area of 615 m²/g. This could be due to carbonisation and activation processes, in which volatile compounds are released as gases and make organic substances unstable due to heat, breaking down molecules' bonds and links [71].

The reference list should be corrected to replace references [54] and [71] from the article. The corrected versions of references [54] and [71], along with their corresponding citations in the sections "2.5. Characterizations of Khat stem activated carbon (Table 1)" and "4.2. Proximate analysis of the KSAC" are provided below:

[54] ASTM International. (1996). ASTM D2854-96: Standard test method for apparent density of activated carbon. West Conshohocken, PA: ASTM International. <https://store.astm.org/d2854-96r04.html>

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The authors apologize for the errors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.